

| Task                                | Method                  | Year | LLM        | Technique                            |
|-------------------------------------|-------------------------|------|------------|--------------------------------------|
| LLM-augmented KG embedding          | Pretrain-KGE [94]       | 2020 | E          | LLMs as Text Encoders                |
|                                     | KEPLER [40]             | 2020 | E          | LLMs as Text Encoders                |
|                                     | Nayyeri et al. [132]    | 2022 | E          | LLMs as Text Encoders                |
|                                     | Huang et al. [133]      | 2022 | E          | LLMs as Text Encoders                |
|                                     | CoDex [134]             | 2022 | E          | LLMs as Text Encoders                |
|                                     | LMKE [135]              | 2022 | E          | LLMs for Joint Text and KG Embedding |
|                                     | kNN-KGE [136]           | 2022 | E          | LLMs for Joint Text and KG Embedding |
| LLM-augmented KG completion         | LambdaKG [137]          | 2023 | E + D + ED | LLMs for Joint Text and KG Embedding |
|                                     | KG-BERT [26]            | 2019 | E          | Joint Encoding                       |
|                                     | MTL-KGC [138]           | 2020 | E          | Joint Encoding                       |
|                                     | PKGC [139]              | 2022 | E          | Joint Encoding                       |
|                                     | LASS [140]              | 2022 | E          | Joint Encoding                       |
|                                     | MEM-KGC [141]           | 2021 | E          | MLM Encoding                         |
|                                     | OpenWorld KGC [142]     | 2023 | E          | MLM Encoding                         |
|                                     | StAR [143]              | 2021 | E          | Separated Encoding                   |
|                                     | SimKGC [144]            | 2022 | E          | Separated Encoding                   |
|                                     | LP-BERT [145]           | 2022 | E          | Separated Encoding                   |
| LLM-augmented KG construction       | GenKGC [96]             | 2022 | ED         | LLM as decoders                      |
|                                     | KGT5 [146]              | 2022 | ED         | LLM as decoders                      |
|                                     | KG-S2S [147]            | 2022 | ED         | LLM as decoders                      |
|                                     | AutoKG [93]             | 2023 | D          | LLM as decoders                      |
|                                     | ELMO [148]              | 2018 | E          | Named Entity Recognition             |
|                                     | GenerativeNER [149]     | 2021 | ED         | Named Entity Recognition             |
|                                     | LDET [150]              | 2019 | E          | Entity Typing                        |
|                                     | BOX4Types [151]         | 2021 | E          | Entity Typing                        |
|                                     | ELQ [152]               | 2020 | E          | Entity Linking                       |
|                                     | ReFinED [153]           | 2022 | E          | Entity Linking                       |
|                                     | BertCR [154]            | 2019 | E          | CR (Within-document)                 |
|                                     | Spanbert [155]          | 2020 | E          | CR (Within-document)                 |
|                                     | CDLM [156]              | 2021 | E          | CR (Cross-document)                  |
|                                     | CrossCR [157]           | 2021 | E          | CR (Cross-document)                  |
|                                     | CR-RL [158]             | 2021 | E          | CR (Cross-document)                  |
|                                     | SentRE [159]            | 2019 | E          | RE (Sentence-level)                  |
|                                     | Curriculum-RE [160]     | 2021 | E          | RE (Sentence-level)                  |
|                                     | DREEAM [161]            | 2023 | E          | RE (Document-level)                  |
|                                     | Kumar et al. [95]       | 2020 | E          | End-to-End Construction              |
| LLM-augmented KG-to-text Generation | Guo et al. [162]        | 2021 | E          | End-to-End Construction              |
|                                     | Grapher [41]            | 2021 | ED         | End-to-End Construction              |
|                                     | PiVE [163]              | 2023 | D + ED     | End-to-End Construction              |
|                                     | COMET [164]             | 2019 | D          | Distilling KGs from LLMs             |
|                                     | BertNet [165]           | 2022 | E          | Distilling KGs from LLMs             |
|                                     | West et al. [166]       | 2022 | D          | Distilling KGs from LLMs             |
|                                     | Ribeiro et al. [167]    | 2021 | ED         | Leveraging Knowledge from LLMs       |
|                                     | JointGT [42]            | 2021 | ED         | Leveraging Knowledge from LLMs       |
|                                     | PSKG2Text [168]         | 2021 | D + ED     | Leveraging Knowledge from LLMs       |
|                                     | GAP [169]               | 2022 | ED         | Leveraging Knowledge from LLMs       |
| LLM-augmented KGQA                  | GenWiki [170]           | 2020 | -          | Constructing KG-text aligned Corpus  |
|                                     | KGPT [171]              | 2020 | ED         | Constructing KG-text aligned Corpus  |
|                                     | Lukovnikov et al. [172] | 2019 | E          | Entity/Relation Extractor            |
|                                     | Luo et al. [173]        | 2020 | E          | Entity/Relation Extractor            |
|                                     | QA-GNN [131]            | 2021 | E          | Entity/Relation Extractor            |
|                                     | Nan et al. [174]        | 2023 | E + D + ED | Entity/Relation Extractor            |
|                                     | DEKCOR [175]            | 2021 | E          | Answer Reasoner                      |
|                                     | DRLK [176]              | 2022 | E          | Answer Reasoner                      |
|                                     | OreoLM [177]            | 2022 | E          | Answer Reasoner                      |
|                                     | GreaseLM [178]          | 2022 | E          | Answer Reasoner                      |
|                                     | ReLMKG [179]            | 2022 | E          | Answer Reasoner                      |
|                                     | UniKGQA [43]            | 2023 | E          | Answer Reasoner                      |